

Identifying Approaches to Encourage Use of Medication IFUs

Introduction

Medication Instructions for Use (IFU) are the primary method for pharmaceutical and combination product manufacturers to communicate important dosing and safety information to patients in the United States. These IFUs are regulated by the United States Food and Drug Administration (FDA) and must strictly adhere to the approved medication labelling and be validated through Human Factors summative testing. This ensures that the IFU will facilitate safe and effective use for the intended user, whether that is a patient, caregiver or healthcare provider (HCP) when they use the IFU. IFUs are only valuable, however, if the user reads them. The goal of this project was to understand why users choose not to read IFUs when first taking a new medication and identify an approach to increase IFU interactions among target users.

Theoretical Background

IFU Literature Research

A great deal of research has been conducted on optimizing IFUs. For pharmaceutical drugs and combination products, the United States Food and Drug Administration (FDA) published a guidance outlining content and format recommendations for developing IFUs (FDA, 2022). The guidance leverages human factors principles for IFU design and provides suggestions on when to bold, avoiding using all capital letters, how to notate warnings and cautions and other style guides. One study evaluated different ways to standardize how to instruct patients to take a multi pill regimen using a patient-centered label that explicitly calls out the time of day for taking the medication (Wolf, et al., 2011). Another study looked at better

ways to organize information on the IFU and alternate ways for folding the IFU to improve searchability for key information (Corderio & Melo, 2021). These are just a few examples of a compelling body of research that exists around designing an effective IFU.

But even the very best IFU is rendered useless if the user does not read it. On this topic, there is comparatively little research.

IFU Formative Research Conducted

In a previous, unpublished project I looked to learn what specific sources of information are preferred by patients for learning about new medication and then seeing if one of those sources, specifically QR codes, could be leveraged to improve the utilization of instructions for oral medication. Following three contextual inquiry interviews, a survey was administered to 62 adult patients. One learning might provide a key insight into why some patients do not read instructions came from the survey. Based on the survey, most patients confirmed that they prefer to receive information from Healthcare Providers (HCPs). This includes information on how to take the medication.

One challenge with relying on the HCPs only to inform patients on how to take their medication is that some patients do not recall the instructions once home and ready to start treatment. In a study that looked at patients going to see their HCP alone due to COVID-19 restrictions, 30% of patients self-reported that they had trouble remembering the instructions from the doctor (Tranos, et al., 2021).

HCPs may also not be clear in communicating the instructions or could provide instructions that are not consistent with the IFU, either intentionally or unintentionally, which could cause patient confusion and lead to medication errors.

In my personal experience running countless Human Factors formative and summative studies for pharmaceutical medication and combination products, I have observed many participants not reading the labelling and IFUs that accompany the medication. The reasons for not reading vary from stating that they rely on information from the pharmacy, which is not verifiable in simulated use testing, to negative transfer of their current mental model for taking medication, to knowing how to take the medication due to something they saw on television or in the media, to many other reasons. However, when patients do not rely on credible sources for instructions, it can create a gap in the path of knowledge from the healthcare experts (manufacturers, HCPs) to themselves and their caregivers. This gap can lead to medication errors. It can also lead to the development of short cuts and work arounds by patients and caregivers that can impact the clinical efficacy of the treatment or lead to a poor user experience.

To bridge the gap of knowledge that is created when users fail to read labelling and IFUs, it is first important to understand their current real-life experiences so that areas of opportunity can be identified and leveraged to improve the likelihood that instructions will be read.

Research Proposal

To better understand the real-life interactions between patients and caregivers and their HCPs, it is important to experience their interactions with each other and the medication in their environments. This proposal looks to conduct contextual inquiry research across the care continuum of a select number of patients and their caregivers.

Study Procedure

This goal of this study is to follow a minimum of 10 patient participants through a cycle of new treatment by HCPs. The study team would first work with recruited HCPs to identify and

recruit patients who would be coming into see the HCP with the goal of starting a new treatment regimen. Prior to the scheduled appointment, the patients and caregivers would be asked to take a survey around how they expect to learn about new medications. The study team would follow the patient participants, some with caregivers and some without, to an appointment with their crazy HCP and observe the interaction between them as the HCP(s) provide instructions on how to take the medication. If the medication is filled via a local pharmacy, that pharmacist would also be recruited if possible, and that interaction would be observed as well. Finally, the study team would be present when the patient takes the medication, with or without their caregiver. The goal of this observation is to track what instructions are given to a patient for a new treatment, in what form, and from whom, culminating in observing how well the patients and caregivers follow these instructions.

A debrief would then be conducted separately with the patients, the caregiver if participating, and any HCPs who were involved in the process. Videos of the initial interactions would be available to view to remind participants of the initial interactions and find the points of communication interactions that may have been missed by the patient, and to determine what drives the patients' and caregivers' understanding of how to take their medication.

Description of Study Participants

This study will have two unique user groups: patients and caregivers of patients. All participants must be 18 years or older. All participants must have opted into participating in the study. Best efforts will be made to recruit participants across a range of literacy, socioeconomic backgrounds, education levels, race and genders. All participants should be able to communicate in English. All participants will not be affiliated with any pharmaceutical or biotechnology company, healthcare company, or government agency that regulates healthcare. HCPs and

pharmacists will be required to disclose any potential conflicts of interest. All participants, whether patient, caregiver or HCP will not have participated in any market research study within six weeks prior to their scheduled session.

Patients

This user group will consist of at least five participants who are seen by five unique physicians. By having five unique patients with five unique HCPs there is a better opportunity to see a variety of different types of interactions, work arounds and interpretations. Patients do not need to be under the care of a specific type of physician, they so need to be prescribed by a qualified prescriber in their state.

Caregivers

This user group will consist of at least five participants who administer medication to a patient in their care. These participants should not be professional caregivers, but rather relations or loved ones who also take on the responsibility of caring for the patient. The patients under the care of these caregivers should see unique HCPs, different from each other and from those physicians treating patients in the patient only arm.

Additional Participant Considerations

While it is ideal for each participant to fill their prescription medication at a unique pharmacy, this is not a requirement for the study.

Ethical Considerations

Minimal Risk

Participants will not be exposed to any risk from materials that were provided by the research team. Patient-participants will administer their own medication prescribed by their physician for the demonstration process as they typically would. To prevent any interruptions to

the participant's regularly scheduled therapy, the research team will not interfere in any way during the administration process.

Ethics

This study will be conducted in accordance with the ethical principles that have their origin in the Declaration of Helsinki and that are consistent with Good Clinical Practice and applicable regulatory requirements. The study will be conducted in accordance with applicable laws, Institutional Review Board (IRB) requirements, and the regulations of the United States FDA as described in Code of Federal Regulations (CFR) 21 Parts 50, 56, and 812 (where applicable).

Data Collection and Privacy

Data will be recorded in the moderator's guide by the moderator. A notetaker will be present supporting as video recording technician during the session. Data will be transcribed into an Excel spreadsheet following a debrief with the research team on-site at the end of each session. In office session, in home sessions, and where applicable and possible, in pharmacy sessions will be video recorded. Analysis will be conducted by aggregating the data obtained to identify themes and common recommendations about the way information is shared with patients and caregivers and what information they then use to administer medication.

Research Team

A maximum of 3 members of the research team will attend each session:

- A moderator to the discussion with the participant.
- A note taker to observe and document the session and ensure the video recording equipment was capturing the interactions. The note taker may also ask additional question needs, as is appropriate throughout the session.

- An observer who may also ask additional question needs, as is appropriate throughout the session.

All members of the research team will be trained in understanding the study protocol prior to the start of any sessions and all researchers will not interfere with a patient or HCP during the office visit or while demonstrating their regularly scheduled therapy during the in-home session.

Withdrawal Criteria

A participant may be withdrawn from the study during or after screening for any of the following reasons:

- Participant does not allow research team to enter their home or office.
- Participant requests the research team to leave their home or office prior to completing the session.
- Participant requests to withdraw from the study.
- Participant does not meet eligibility criteria as outlined in the screener.
- Participant does not complete the series of questions at the beginning of the study.
- Based on moderator's observation, participant does not understand the study procedure and/or is in danger of causing harm to self or others.

Capabilities Assessments

At the end of each session the moderator will administer the Newest Vital Sign (NVS) health literacy functional capability assessment to all patient and caregiver participants (Kunter, et al., 2006).

The NVS is a test using a nutrition label that is accompanied by six questions and requires approximately three minutes for administration. Patients with four or more correct responses are unlikely to have low literacy, those with two to three correct responses possibly have limited literacy, and those with zero to one correct response have a high likelihood (50% or more) of limited literacy. Limited health literacy has been shown to be a potential risk factor that might lead to higher rates of outpatient medication errors because of improperly following dosing administration instructions. Health literacy is the “degree to which individuals have the capacity to obtain, process and understand basic health information and services need to make appropriate health decisions” (Ratzen & Parker, 2000).

How the Proposed Research Could Encourage IFU Usage

This research will serve to identify key insights into why patients and caregivers do not use IFUs, and how their preference for learning from HCPs informs their understanding of the proper way to take medication. By better understanding the real-world interactions with HCPs, and observing what patients and caregivers remember, what they do not remember, and what they do if they realize that they do not remember, current IFUs can be modified or completely changed to better compliment treatment initiation, whether for the patient and caregiver, or for the HCPs.

Innovations and Forecasts of Future Trends

One of the goals of this research is to develop new innovations in IFUs that will compel patients and caregivers to want to use them as they take medications. Following the observations sessions, there is the potential to discuss ideas for innovating on the current IFU designs seen with their medications. Potential ideas that can be explored during the debrief sessions of the study could include the value of shorter, quick start guides and how to disseminate them,

manufacturer training for HCPs with corresponding information for the HCP to share with the patient, or any number of other ideas that will come directly out of the observational contextual inquiry research. Incorporating digital IFUs or alternate formats for IFUs such as largely pictorial instructions or small booklets could better meet the needs of the user.

Other future trends could possibly include the HCP in the dissemination of information process to leverage the trusted source of those individuals. Manufactures could look to provide annotated copies of the IFUs for HCPs to use when training their patients and their caregivers. This would allow for the patients and caregivers to receive explanations on how to take their medication from their HCP but then still have the IFU document that they receive in the medication package be consistent with that used by their HCP. Regulatory limitations could hinder the implementation of new innovations. As with any innovation getting users to buy into something novel can be a limitation as well. However, depending on the outcomes of the research and future design, the benefits could far outweigh the hurdles in leading to better patient safety and user experience.

References

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